

This project asks the candidate to use any kind of mathematical and statistical models to study the various data sets' forecast power of future security returns. The end goal is to build a profitable portfolio with consistent risk adjusted return based on the data sets and reference data provided here. An ideal portfolio would have a high Sharpe ratio with low drawdown (peak to trough). All the data are provided in the format of csv files.

NOTE: Please **do not** use any hardware or software or IP that belongs to your current employer or institution in completing this assignment. Please **do not** seek advice or help from other individuals. This should be your own work product.

Security reference data

The names of security reference data files are "security_reference_data_w_ret1d_1.zip" and "security_reference_data_w_ret1d_2.zip". Columns included in the file: "data_date", "security_id", "ret1d", "close_price", "volume", "group_id", "is_tradable".

"data_date" is the date when the security is traded.

"security_id" is a unique id associated with each security.

"ret1d" is the price return. For security i at trading day t , the 1 day return is given, $ret1d(i,t)$. This return is price return from trading day $t-1$'s close price to trading day t 's close price. If $ret1d(i,t)$ has a value $+0.1$, it means security i 's close-to-close return on trading day t is $+10\%$.

"close_price" is the close price in \$ when market closes, i.e. last price of the day.

"volume" is total amount of shares traded for the security on that day.

"group_id" is an 8-digits-integer and identifies which group the security belongs to. The first 2 digits identify the level I grouping, the first 4 digits identify level II grouping, the first 6 digits identify level III grouping and the whole 8 digits identify level IV grouping. E.g. for $group_id=10503060$, the level I group is 10, level II group is 1050, level III group is 105030, level IV group is 10503060. It is at candidate's discretion whether to use any of this.

"in_trading_universe" is a flag ("Y" or "N") to indicate whether this security is eligible to be included in the portfolio.

It can be assumed here all the values here are available after the market close on the "data_date".

Data sets

Several data sets are provided here, label as d1, d2, d3 and etc. Each data set is stored in a separate file, labeled as "data_set_1.csv" or "data_set_2.csv" and etc. There are 3 columns: "data_date", "security_id", and a value column which could be "d1", "d2" and etc. "data_date" is the date when the value column is available. It can be assumed the value column is available after the market close on the "data_date".

Risk factors

A few risk factor exposure scores are provided. They are all stored in the files named "risk_factors_1.zip" and "risk_factors_2.zip". For every risk factor, it has an exposure score for each security at each trading

day. The exposure score can be thought as a measure of how much a security is exposed to the volatility of risk factors. Column "rf1" contains the exposure scores of risk factor rf1 for all the securities across different dates. The column "data_date" is the date when exposure scores are computed. It can be assumed that exposure scores are available after the market close on the "data_date".

It is at candidate's discretion whether to use any of the risk factors here for the project.

Portfolio construction, trading and return calculation

The target portfolio should be market neutral, i.e. absolute position notional of long side book should be same as absolute position notional of short side book. Assuming position weight for security i at the market close of trading day t is $w(i,t)$, $\sum(w(i,t))=1$ for all the long names with $w(i,t)>0$ and $\sum(w(i,t))=-1$ for all the short names with $w(i,t)<0$. Equivalently, portfolio can be thought to have long book size of \$1 and short book size of -\$1.

The portfolio constituents are limited to the securities which are in the trading universe (aka `in_trading_universe="Y"`). The number of securities which are held by the portfolio at any given day is up to the candidate. E.g. while trading universe has 2000 securities at trading day t , the portfolio could hold all OR a subset of these 2000 securities. In the case that a security drops out of trading universe (`in_trading_universe="N"`), it needs to be liquidated from the portfolio.

Assuming portfolio rebalances once a day and only trades at the close of the market. Here the trading cost can be assumed to be 0.01% of trading notional (i.e. if you trade \$1 worth of security, your one way trading cost is 0.01 cent) and the execution price is exactly the "close_price". Thus realized daily return of the portfolio on trading day $t+1$, $\text{port_ret}(t+1)$ can be simply calculated as $\text{port_ret}(t+1) = \sum(w(i,t)*\text{ret1d}(i,t+1))$ minus the trading cost.

It is important to pay attention to the time sequence at which data becomes available when building your alpha model and portfolio.

Deliverable

- 1) A copy of final project report (not including source code) is expected to be no more than 10 pages.
 - a. Describe your intuition and reasoning used behind steps of data processing, model building and portfolio construction. Some example questions to be answered: why you choose to use certain data sets, what make you pick the certain model over others and etc.
 - b. Plot and analyze the portfolio performance. List out all the performance statistics: return, volatility, drawdown (magnitude and length), Sharpe ratio, % of positive days, turnover and etc. A time series plot of portfolio cumulative return should be presented. Feel free to add any graph you think is useful. The cumulative return shall be calculated by using arithmetic daily portfolio return (assuming portfolio book size remains the same).
 - c. Discuss the impact of trading cost and rebalance frequency on your portfolio.
 - d. Discuss future potential improvement and concerns if any.
- 2) A copy of the source code which you used to process the data, generate forecast, construct portfolio (Python notebook preferred). Along with the instruction to run the code.